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openQSE - Q Control WG

Sep 23, 2026

Andrea Delgado yasuko.eckert@amd.com Daniel Weigand aaron.lott@hpe.com
matthias@mq.sc daniel@mq.sc josem.monsalvediaz@amd.com nunezcorrals@ornl.gov
shehataa@ornl.gov langejr@ornl.gov michael.ferguson@hpe.com ~~Fokke de Vries~~

Group organized white paper goals with architecture scopes via milestone planning.

Working Group Governance and Milestone Objectives

- Andrea Delgado stated that working groups require co-chairs from both vendors and national labs.
- Jack Lange joined Andrea Delgado to co-chair the quantum control working group.
- Andrea Delgado outlined the first milestone to produce a white paper by SC detailing requirements.
- Michael Ferguson noted the white paper should establish community standards for control electronics programs.

Functional Scope and Architecture Boundaries

- Andrea Delgado proposed focusing the scope on on-premise execution rather than cloud access models.
- Michael Ferguson agreed that resource management scheduling falls outside this working group domain.
- Santiago Nunez-Corrales proposed distinguishing high-performance computing scheduling from micro-architecture scheduling.
- Matthias Traube suggested limiting safety features to preventing users from damaging QPUs with pulses.

Distributed Quantum Systems and Entanglement Requirements

- Santiago Nunez-Corrales proposed including distributed quantum systems only when managing entanglement at control levels.
- Amir clarified that deployments focus on entanglement rather than parallel multi-QPU setups.
- Santiago Nunez-Corrales highlighted circuit cutting and knitting as use cases requiring real-time synchronization.

Terminology Alignment with System Architecture Group

- Amir noted that the system architecture working group is drafting a terminology white paper.
- Andrea Delgado initiated cross-group representation to ensure terminology cohesion with the system architecture team.

Control flow and qubit scale requirements

- Systems with thousands of qubits require QC information to pass directly to electronics.
- Andrea Delgado stated that the group does not want to define control flow requirements currently.

Working group documentation and repository management

- Michael Ferguson noted that meeting notes reside in a branch rather than the main fork.
- Amir confirmed granting admin access to push the meeting notes to the main repository.

Next steps

- ☐ [Andrea Delgado] Distribute Welcome Deck: Distribute the welcome slide deck containing scope, milestones, and goals for the working group to all members.
- ☐ [The group] Review Meeting Deliverables: Review the posted meeting deliverables on GitHub or Slack and provide feedback or suggestions for improvement.
- ☐ [Santiago Nunez-Corrales] Document Scheduling Distinctions: Clarify the differences between High-Performance Computing scheduling and micro-architecture scheduling requirements for the control stack in a written document.
- ☐ [Amir] Send Working Group Contact: Provide contact information for the System Architecture working group leads to facilitate better coordination.
- ☐ [The group] Vote and argue on use cases: Evaluate proposed use cases and provide feedback on whether to include them.
- ☐ [Andrea Delgado] Share use case documentation: Distribute the use case table and template to all participants.
- ☐ [Andrea Delgado] Compile use case feedback: Aggregate the voting and arguments from the group by next Wednesday.
- ☐ [Andrea Delgado] Push notes to repository: Upload existing notes to the main working group controls fork.
- ☐ [Andrea Delgado] Share GitHub link and notes: Provide the repository access link and post the meeting transcript summaries.

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Invited [Andrea Delgado](#) yasuko.eckert@amd.com [Daniel Weigand](#)
aaron.lott@hpe.com matthias@mq.sc daniel@mq.sc
josem.monsalvediaz@amd.com nunezcorrals@ornl.gov shehataa@ornl.gov
langejr@ornl.gov michael.ferguson@hpe.com ~~[Fokko de Vries](#)~~

Attachments  [openQSE - Q Control WG](#)

Meeting records  [Transcript](#)

Summary

Group organized white paper goals with architecture scopes via milestone planning.

White Paper Planning

Milestone targets Supercomputing conference for landscape mapping. Fault tolerance scopes hybrid execution use cases.

Architecture Boundaries

On-premises execution focuses from pulse level down. Distributed quantum computing remains in scope for entanglement.

Use Cases Assignment

Participants voted on use cases. GitHub repository access was established.

Decisions

Aligned

- **Scope limitation for cloud access and scheduling** The working group aligned to focus exclusively on on-premises execution and runtime-to-control feedback boundaries while excluding cloud access and high-level scheduling.
- **Inclusion of distributed quantum computing use cases** The working group aligned to include distributed quantum systems as a speculative use case restricted to scenarios involving entanglement managed by the control stack.
- **Exclusion of fine-grained security requirements** The working group aligned to exclude fine-grained facility-scale security and tenancy requirements from milestone one, retaining only basic instruction safety safeguards.

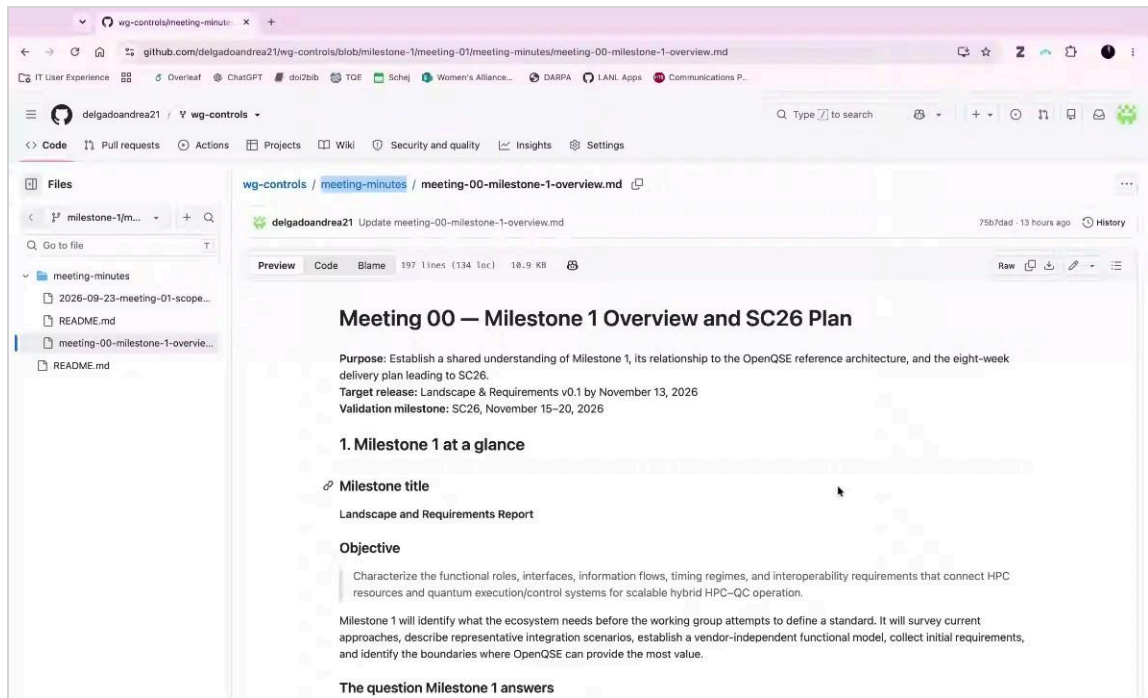
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Details

- **Working Group Organizational Structure and Co-Chairs:** Andrea Delgado opened the bi-weekly meeting of the Open QSC quantum control group by announcing that because Open QSC is funded by the Department of Energy, working groups must be co-chaired by representatives from both a vendor and a national lab. Jack Lange was introduced as a co-chair representing Oak Ridge National Laboratory, joining other participants who introduced themselves, including Santiago Nunez-Corrales from Oak Ridge National Laboratory, Daniel Haag from MQSC and the Technical University of Munich, Daniel Weigand from Cube Blocks, Michael Ferguson from Hewlett Packard Enterprise, and Matthias Traube from MQTT ([00:00:20](#)).
- **White Paper Milestone and Supercomputing Conference Target:** Andrea Delgado outlined the first major milestone, which is producing a white paper targeting the Supercomputing conference to map the quantum control landscape, interface requirements, ecosystem needs, and establish a vendor-independent functional model. Michael Ferguson and other participants noted that the white paper will not serve as a normative specification, will not endorse a preferred vendor or software stack, and will not establish performance benchmarks or device-specific integrations during this initial phase ([00:04:03](#)) ([00:08:27](#)).



- **Reference Architecture Integration and Deliverables:** Andrea Delgado discussed planned deliverables and use cases to be hosted on the Open QSC GitHub repository, noting that Amir aims to showcase a reference architecture demonstration by the Supercomputing conference ([00:05:29](#)) ([00:09:58](#)). Andrea Delgado proposed using this reference architecture as a feedback process rather than a direct starting point for the white paper, ensuring that the working group can evaluate contributions before defining future milestones ([00:11:23](#)).
- **Fault-Tolerant Quantum Computing and Hybrid Execution Scope:** Santiago Nunez-Corrales raised a question regarding whether Fault-Tolerant Quantum Computing requirements should be implicit or explicit. Andrea Delgado and Santiago Nunez-Corrales agreed that while fault tolerance is assumed for accelerator interfaces, the team can carve out specific hybrid execution use cases that do not immediately require full fault tolerance to map out hybrid interfaces effectively ([00:12:47](#)).

github.com/delgadoandrea21/wg-controls/blob/milestone-1/meeting-01/meeting-minutes/2026-09-23-meeting-01-scope-charter.md

milestone-1/mee... wg-controls / meeting-minutes / 2026-09-23-meeting-01-scope-charter.md

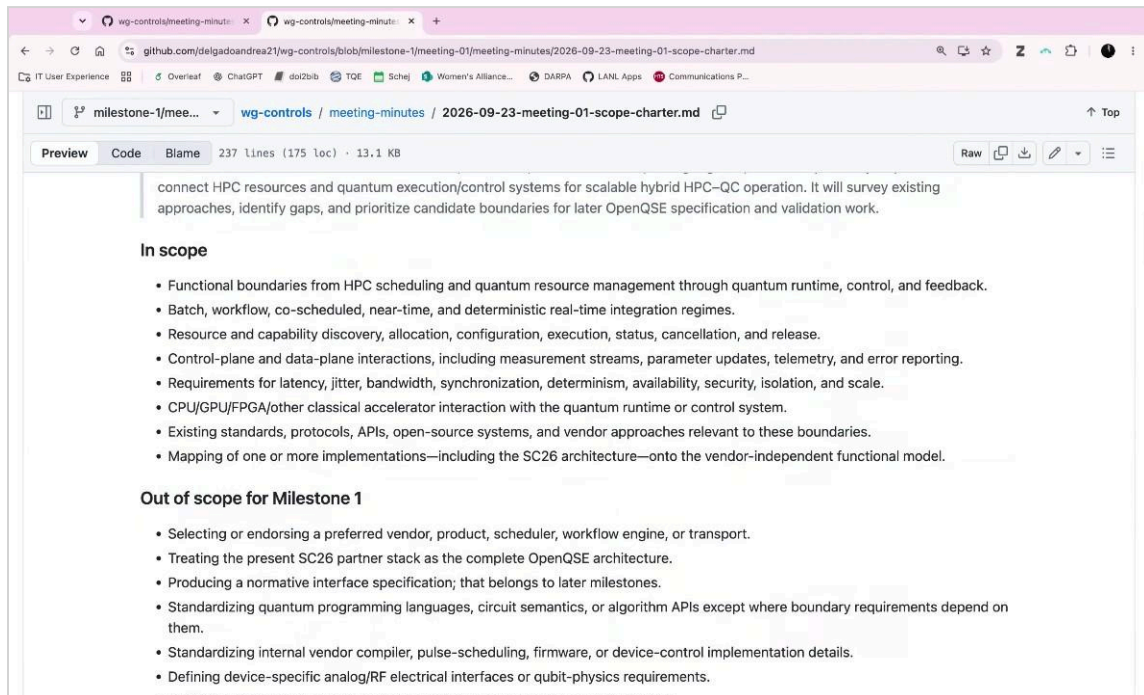
Preview Code Blame 237 Lines (175 loc) · 13.1 KB

The use cases are requirement drivers, not commitments that OpenQSE will standardize every interface they touch.

ID	Use-case class	Short scenario	Timing regime	Proposed relevance
UC-01	Remote or batch quantum execution	An HPC job submits work to a quantum resource and later retrieves status and results	Loose / asynchronous	Context
UC-02	Workflow-level hybrid execution	A workflow coordinates dependent CPU, GPU, and QPU tasks with data exchange	Loose to near-time	Adjacent
UC-03	Co-scheduled HPC-QPU execution	CPU/GPU/QPU resources and network paths are reserved together for one hybrid job	Scheduling-sensitive	Primary
UC-04	Iterative calibration and adaptive execution	Measurement results drive parameter updates or subsequent executions within a session	Near-time	Primary
UC-05	Deterministic classical-to-control interaction	Classical compute exchanges data or commands with control hardware under bounded timing	Real-time	Primary
UC-06	QEC decoding and fast feedback	Measurements stream to a decoder and corrections or frame updates return before a deadline	Hard real-time / streaming	Primary
UC-07	Multi-QPU or distributed execution	A workflow coordinates multiple QPUs or control domains with topology and synchronization constraints	Mixed	Discussion needed

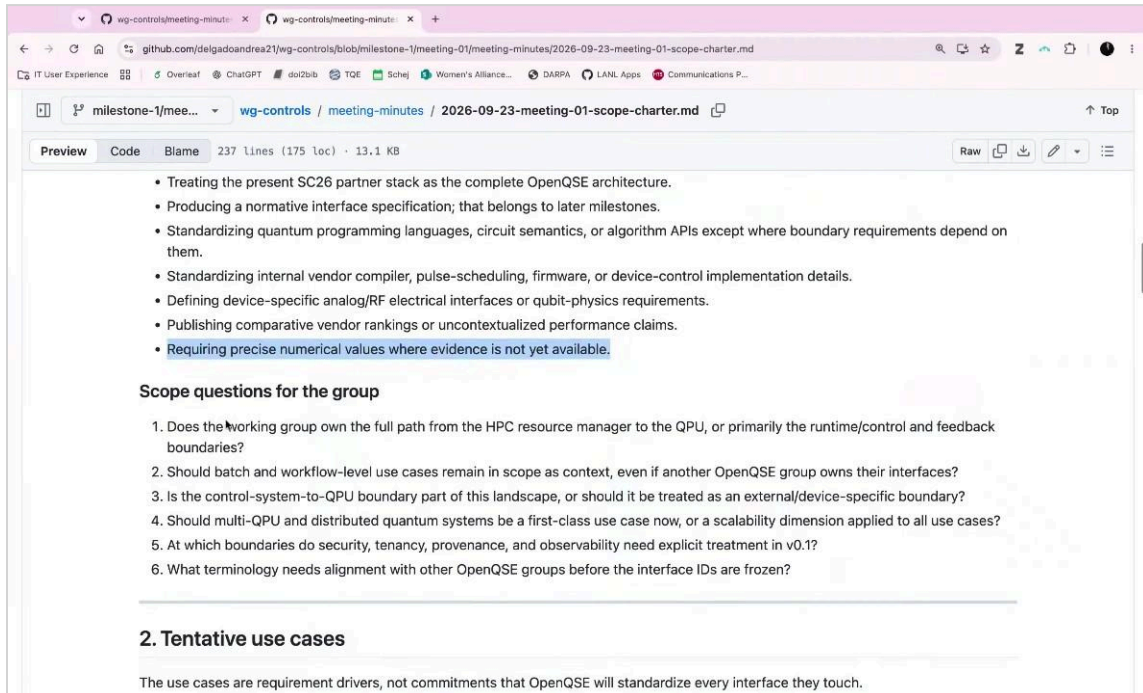
Prioritization questions

- Functional Boundaries and On-Premises Execution Focus:** Daniel Weigand questioned the level of detail and functional boundaries needed regarding resource management and scheduling (00:14:58) (00:19:30). Michael Ferguson argued that high-performance computing scheduling is outside the working group's domain and that discussions should focus on on-premises execution from the pulse level down, assuming that co-scheduled real-time decoding or feedback resources are located locally next to the quantum processing unit rather than accessed via the cloud, with Matthias Traube supporting this view (00:21:13).
- Compiler Interfaces and High-Performance Computing Scheduling Distinction:** Daniel Weigand highlighted the missing interface to the compiler group regarding gate-level to pulse-level compilation, caching, and partial compilation (00:22:56). Michael Ferguson explained that control electronics' program requirements must first be defined (00:24:15). Additionally, Santiago Nunez-Corrales distinguished between traditional high-performance computing scheduling such as Slurm and the immediate, regular, fast access required to support microarchitecture runtimes, committing themselves to write about this distinction (00:25:41).



- **Telemetry, Data Volumes, and Out-of-Scope Items for Milestone One:**

Daniel Weigand and Santiago Nunez-Corrales discussed system administration, hardware failure monitoring for electronics, data volume, program sizes, and communication intensity to identify bottlenecks. Andrea Delgado confirmed these elements fall under telemetry and classical accelerator interfaces (00:28:36). Furthermore, Andrea Delgado reiterated that vendor product endorsements, device-specific interfaces, qubit-agnostic bottlenecks, and precise numerical values are out of scope for milestone one and are deferred to later milestones (00:30:28).



- **Distributed Quantum Computing and Multi-Quantum Processing Unit**
Scope: Addressing multi-quantum processing unit deployments, Santiago Nunez-Corrales, Michael Ferguson, and Amir debated whether distributed quantum computing should be included ([00:32:02](#)). Amir initially felt it was too speculative, but Michael Ferguson and Santiago Nunez-Corrales argued that quantum networking, entanglement, and circuit cutting or knitting require low-latency communication paths and soft real-time synchronization ([00:33:58](#)) ([00:39:15](#)). The group reached a consensus led by Andrea Delgado and Amir that distributed quantum computing is in scope only when it involves entanglement managed by the control stack, whereas embarrassingly parallel jobs without entanglement belong to higher scheduling levels ([00:32:02](#)) ([00:36:35](#)).
- **Security Features and Hardware Protection Limits:** Andrea Delgado questioned whether security tenancy and high-performance computing facility requirements should be considered. Michael Ferguson stated these details are too fine-grained for milestone one. Matthias Traube suggested that the only necessary security feature is ensuring users cannot send instructions that physically destroy or fry the quantum processing unit with pulses, though Daniel Weigand noted this might be difficult depending on the specific qubit architecture ([00:40:56](#)).

boundaries?

2. Should batch and workflow-level use cases remain in scope as context, even if another OpenQSE group owns their interfaces?
3. Is the control-system-to-QPU boundary part of this landscape, or should it be treated as an external/device-specific boundary?
4. Should multi-QPU and distributed quantum systems be a first-class use case now, or a scalability dimension applied to all use cases?
5. At which boundaries do security, tenancy, provenance, and observability need explicit treatment in v0.1?
6. What terminology needs alignment with other OpenQSE groups before the interface IDs are frozen?

2. Tentative use cases

The use cases are requirement drivers, not commitments that OpenQSE will standardize every interface they touch.

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UC-01	Remote or batch quantum execution	An HPC job submits work to a quantum resource and later retrieves status and results	Loose / asynchronous	Context
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UC-03	Co-scheduled HPC-QPU execution	CPU/GPU/QPU resources and network paths are reserved together for one hybrid job	Scheduling-sensitive	Primary
UC-04	Iterative calibration and adaptive execution	Measurement results drive parameter updates or subsequent executions within a session	Near-time	Primary

- Terminology Alignment with System Architecture Working Group:** Andrea Delgado and Amir discussed terminology alignment regarding concepts like compilation and timing domains such as real-time versus near-time. Amir noted that the system architecture working group, run by Johannes Belchuk and Sebastian Stern, is drafting a white paper on terminology, prompting the group to coordinate and exchange representatives to ensure cohesion across the Open QSC initiative (00:42:38).
- Defining Control System and Quantum Processing Unit Boundaries:** Daniel Weigand, Michael Ferguson, Jack Lange, Santiago Nunez-Corrales, and Andrea Delgado engaged in an extensive debate on how to define the quantum processing unit and the control system (00:43:53). Jack Lange proposed defining the control system as non-quantum infrastructure co-scheduled with quantum infrastructure, while Michael Ferguson and Santiago Nunez-Corrales framed it as dedicated system-level hardware or a virtualized microarchitecture translating symbolic program instructions into physical hardware actions (00:48:18). Andrea Delgado emphasized that control boxes also perform information processing and data computation during readout, and Jack Lange expressed concern about scope creep regarding how deeply the control system interfaces with quantum error correction decoders and compiler teams (00:53:40) (00:57:25).

Opening Up the Stack

LEGEND

- NISQ-RT
- QEC
- Physical Qubit Level Info
- QCS

The diagram illustrates the flow of information and processing in the 'Opening Up the Stack' architecture. It starts with a **User** providing input to the **LOGICAL CIRCUIT ALGO**. The **LOGICAL CIRCUIT ALGO** includes **QEC injected circuitry** and **QEC injected circuitry (QECs)**. The output of the **LOGICAL CIRCUIT ALGO** is then processed by the **Instructional Parameters** block, which includes **Physical Qubit Parameters**, **Data Parameters**, and **Error Rate Model**. The **Instructional Parameters** block outputs to the **Pre-compiled Instructions** block. The **Pre-compiled Instructions** block includes **Instructional Parameters**, **Acquired Data**, and **Decoder**. The **Pre-compiled Instructions** block outputs to the **Computer Core**, which includes **Calculating Instructional Parameters** and **Decoder**. The **Computer Core** outputs to the **Decoder**.

Inset: LOGICAL CIRCUIT ALGO

The inset shows a detailed view of the **LOGICAL CIRCUIT ALGO**, including a table of **Physical Qubit Parameters** and a **Physical Qubit Level Info** table.

Qubit ID	Qubit Type	Qubit Frequency	Qubit Error Rate	Qubit Connectivity
1	Qubit 1	1.0	0.01	1
2	Qubit 2	1.0	0.01	2
3	Qubit 3	1.0	0.01	3
4	Qubit 4	1.0	0.01	4
5	Qubit 5	1.0	0.01	5
6	Qubit 6	1.0	0.01	6
7	Qubit 7	1.0	0.01	7
8	Qubit 8	1.0	0.01	8
9	Qubit 9	1.0	0.01	9
10	Qubit 10	1.0	0.01	10
11	Qubit 11	1.0	0.01	11
12	Qubit 12	1.0	0.01	12
13	Qubit 13	1.0	0.01	13
14	Qubit 14	1.0	0.01	14
15	Qubit 15	1.0	0.01	15
16	Qubit 16	1.0	0.01	16
17	Qubit 17	1.0	0.01	17
18	Qubit 18	1.0	0.01	18
19	Qubit 19	1.0	0.01	19
20	Qubit 20	1.0	0.01	20

- **Control Flow and Requirements for Large-Scale Qubits:** Daniel Weigand raised a technical concern regarding how information for large-scale cubes consisting of thousands of qubits must be passed to their electronics as a single package due to program storage limitations. Jack Lange questioned whether their group should formally define this control flow or focus on gathering requirements. Andrea Delgado concluded that the group should not define anything at the current stage to respect the meeting schedule, directing their focus instead to upcoming use cases ([00:58:34](#)).
- **Use Cases Homework Assignment and Next Meeting Timeline:** Andrea Delgado assigned homework asking participants to vote and argue over the next couple of days on whether to include specific use cases from a shared table. Andrea Delgado committed to compiling this information by next Wednesday and soliciting volunteers to dive deeper into use cases and requirements. Andrea Delgado also stated they will share a template of standardized questions to streamline the process, with the goal of agreeing on these use cases during the meeting scheduled in two weeks. Santiago Nunez-Corrales expressed excitement about the engagement, a sentiment which Andrea Delgado echoed ([00:59:49](#)).
- **GitHub Notes Repository Access and Asynchronous Communication:** Michael Ferguson inquired about the location of shared meeting notes, pointing out that they were in a specific branch rather than the main fork of working

group controls, and asked if Andrea Delgado planned to push them to the main repository. Andrea Delgado replied that they had already submitted a pull request pending approval, while Amir noted that they had granted Andrea Delgado admin access to push directly. Andrea Delgado agreed to share the GitHub link, instructing participants to contact either Andrea Delgado or Amir if access issues arose, and designated Slack as the primary platform for asynchronous discussion where they would post the transcribed notes ([00:59:49](#)).

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openQSE - Q Control WG - Transcript

00:00:20

Santiago Nunez-Corrales: Hello.

Andrea Delgado: who's joining from the C account. Michael,

openqse: Hello.

Andrea Delgado: I we were just confused cuz your name is Open QSC with the Open QSC logo.

openqse: You know, I'm not sure how I managed that. Oh, I know. I know why. I know why. It's because I was logged in as open QSC to post some things to YouTube.

Andrea Delgado: I see.

openqse: Maybe I will log out and come back so that the transcript is not very confusing.

Andrea Delgado: That's fine. So for the sake of time, we have a very packed agenda today. Um I know we haven't been meeting but um some of us have been working on the background to get things moving. So I know that Amir is supposed to join and ideally he will be already here but let's just go ahead and get started. Um so again this is a bi-weekly meeting of the open QC quantum control uh group. We have a few organizational announcements. Um ideally since open QSC is funded by DOE and it makes sense that some of the work that we do since most of us are investing like personal or like work time on this that um every single working group is co-chaired by someone who comes from like a vendor in a national lab.

00:02:49

Andrea Delgado: Um so we have asked Jack to join us in co-chairing this working group and we also have a few new faces joining the group today. So if you don't mind very quickly just going around and like just saying like who you are like um where you work basically and just keep it short because we have a very packed agenda today. So I'm gonna start with Santiago because you're the first one on my screen.

Santiago Nunez-Corrales: Um, very nice to meet you all. Santiago Nunes Corales. I work at Oakridge National Lab and I serve as the lead quantum HPC scientist.

Andrea Delgado: Good. We have two Danos, but I'll start with the Daniel that I don't know.

Daniel Haag: Uh, I guess that's me. Uh, I'm Daniel. I am a software engineer at MQSC and also a PhD student at the Technical University of

Andrea Delgado: Awesome.

Daniel Haag: Munich.

Andrea Delgado: Uh, Jack,

Jack Lange: Uh Jackum I work at Oakridge um in the National Center for Computational Science um and I'm a project lead on the QC

00:04:03

Andrea Delgado: um, Daniel from Cube Blocks.

Daniel Weigand: Yeah. Hi, I am Daniel. I am a technical project manager for all things software. So, uh also joining Andrea

Andrea Delgado: Good Michael.

Michael Ferguson: Hi, I'm Michael Ferguson. I'm from HPE where I'm the software architect on the quantum team. I'm also the compiler working group leader for Open QSC.

Andrea Delgado: Um, and last but not least, Matias

Matthias Traube: Uh, hello. I'm Matias. I'm the head of product at MQTT.

Andrea Delgado: Okay. Um, awesome. So, uh, since some of you are new, um, I will try to send after this meeting, um, sort of like the welcome slide deck that we have in terms of like scope, milestone, goals for this working group. Um but the idea is to go from understanding the landscape of uh quantum control and specifically how it interfaces with um the software ecosystems that are developing and the discussion of course is along the lines of open interfaces and standards. Um I think for the quantum control um I guess part of the stack is a little bit of um an interesting discussion to have.

00:05:29

Andrea Delgado: So we have I guess a bigger challenge than the rest of open QSC. Um so the first milestone that we have is um like this white paper where we compile information about um where do we sit in like just like a like the full stack I guess architecture within Open QSC. Um how does it interface? Um we had this diagram of um information flow that we discussed during our first on-site meeting during open QSC day. And the idea will be basically to translate that information diagram into a white paper and um a list of like interfaces and requirements in terms of like latency, bandwidth and and all of that. So I prepare a program to do basically a sprint towards SC to have that white paper in place by by SC. So everything that I'm going to be showing from now on is going to leave on the GitHub repo for Open QSC. So if you don't have access to this repo, please ping me on Slack uh after the meeting and I can give you access so you can edit and and also view um a lot of these things.

00:06:59

Andrea Delgado: The meeting notes would also be posted there. Um I'm going to be compiling information in a certain way so we can have like a standard format. So

everything from now on will leave on on this GitHub repo. And the idea is also that the other working groups will also share this this repo.

Jack Lange: Oops.

Andrea Delgado: Uh so again this lives within open QSC. Is this visible? Okay. Or should I zoom in?

Michael Ferguson: It's fine.

Andrea Delgado: Okay. Um so within um meeting minutes I created basically like an agenda for now up to SC where I'm hoping we can have a first draft of or white paper and I'll walk you through the motivation of um why SC. So again the first milestone is this white paper where we have like the landscape and and requirements. Um identify the ecosystem needs. Um and I think this is important because when we were having the first discussion before we talk about like standards we need to understand what these different interfaces are and also uh as we mentioned establish like this vendor independent uh functional model because of course if we think among the different architectures that we have for cubits every single one of these will have different requirements in terms of like data transfer how they interface with the software stack.

00:08:27

Andrea Delgado: So in a general picture um this white paper or this milestone um as we call it ideally it will answer like what interfaces are required for scalable interoperable HPC quantum integration and where do the existing approaches leave um any gap. I want to clarify that this white paper is not a normative specification. It's we're not trying to like set um a reference that everybody should like start working on. Um of course uh within Open QSC we don't have a preferred vendor of product like this should be as vendor agnostic as possible. We are also not endorsing one software stack or or architecture. Um I know higher up in the stack we have QDMI, we have QRMI. We'll try to integrate with as many um software stacks and architectures as possible. And as of now um if you go back to the other milestones that we have as of now the goal is not to establish performance benchmarks. Um we're not trying to integrate into a specific system. This is just gathering information to understand where we are and where the gaps are and also understanding um the different latency I guess

um or I guess where do we sit in like the different timing domains that we work in on the software stack.

00:09:58

Andrea Delgado: So I'm thinking that for milestone one we can also not only have the white paper as a product but we can start compiling a set of uh use cases and this is um independent of the white paper a functional architecture that we can eventually like um use for future reference. Um, and I have a few other deliverables. So, I'm not going to go through this because it's a lot. But if you have well please allocate time to go through this um after the meeting and just comment directly or either on GitHub or on Slack if you have any um questions uh suggestions about um this mapping and I don't know how many of you were on the Monday like open QC general meeting but The goal of Amiris actually having this open QSC reference architecture. So what this means is he's looking to have um sort of like a demo or like um a showcase of the open QSC architecture by SC and this is what the larger Open QSC group has in mind and it goes all the way down to the control stack.

00:11:23

Andrea Delgado: So as of now I think it's just too early to start integrating into this reference architecture but I think we can start thinking about how we can integrate or how can we connect with this um this reference architecture. So I see or the way I'm seeing this is we should not take this reference architecture as um as a starter for our discussion point for our white paper and the set of deliverables that I proposed that we have but we sort of have like a feedback That's process with the reference architecture. Is this something that we already see in the open QSC reference architecture? Is there something missing? Is there something that we can actually contribute? And then um as we complete milestone one, of course, this will flow into how we work for the the other three milestones that we have as of now. Any questions

about the plan? I'm hoping that for this um meeting we can define the scope and identify uh stakeholders for the different pieces of the paper.

Santiago Nunez-Corrales: Thank Andrea.

00:12:47

Santiago Nunez-Corrales: Uh do we leave FTQC implicit or explicit requirements for

Andrea Delgado: Um, when you say implicit or explicit, what do you mean?

Santiago Nunez-Corrales: um yes what I mean is that they are not FTQC is not explicitly stated across what you have presented. Should we immediately assume that it is toward FTQC or do we have a bridge in between to fill

Andrea Delgado: Um from the initial discussions that we had uh the idea was that we always consider falter and and I think um especially

Santiago Nunez-Corrales: Okay.

Andrea Delgado: for the control stack layer it makes sense because that's where the different um interfaces with accelerators um like GPUs in in classical nodes make sense. But to your point, um, so I do have the way I'm seeing this paper going and I'll go through details in in a bit. I'm thinking that we as a first uh step towards this white paper, we start identifying use cases and I'm thinking we can pick like between three or five and I've listed here a few use cases. So I think if we like most of these are like on the fall tolerance side but like I think if we focus on hybrid execution for example we could carve out maybe a space that doesn't involve fully like fall tolerant and we just map out a use case where we have like a hybrid interface

00:14:58

Santiago Nunez-Corrales: Yep. Thank you. This makes a lot of sense.

Andrea Delgado: Daniel.

Daniel Weigand: um what what's the level of detail we're aiming for here? Because in the in the general architecture, I think the entire working group is basically one box. Um do we then break out that box and specify interfaces within

Andrea Delgado: Um that's one of the questions. So I I I prepare a set of questions to define um the scope for this um white paper and that's actually the first question that I have like at which point should we start considering the work of other working groups like for example like resource management like um should we only start from how they interface um or like how information gets sent from like pulse level down or for the use case we need to include perhaps how those resource get allocated. So that's indeed one of the discussion points for today. Okay, since we have a lot to discuss, um, so we have now sort of like a general overview of like open QSC, SC reference paper.

00:16:37

Andrea Delgado: Um, I have an agenda here of um, work and decisions that need to be made um, to make this uh, white paper happening. So for today again I would like to discuss like the general scope like we don't have to go into many details just like high level this is in scope this is out of scope um this is where we're placing boundaries with the rest of the of the open QC working group um ideally we can select some use cases and again I'm hoping that we start with a use case set regardless of the white paper something that we can reuse for later. Um some interface taxonomy so we all start speaking the same language. Um a technology landscape. So we don't have uh many contributions from the control side but perhaps we can involve uh the labs the open source um initiatives that we have um and the full stack u members that that we have and then again this discussion of like how we feed with um the open QSC reference architecture. Uh so yeah um this is what I I wanted to highlight as well.

00:18:06

Andrea Delgado: I think um we could organize ourselves into five different like subworking groups or if you want to if you have the bandwidth to work in more than one of these work streams it would be great. Um one of them as I mentioned is uh the scope and and use cases mirror is here. The other one is architecture in and interfaces. Um what else is back the technology landscape uh requirements and gaps and also how do we interface with the SC26 um integration again um more metrics for success. But I think this is just something that you can read on your own um before the next meeting. Okay. Um so for this meeting I would like to leave this meeting with um a scope for what we want to include in in this white paper um a decision on which use case uh we should have. I show you like a a strollman diagram of how we can see the entire software stack without going into like details and if we should add remove uh perhaps identify like different sidelines for this architecture.

00:19:30

Andrea Delgado: But this would serve as a baseline as we start thinking about the different um interfaces um and also assigning owners for all of these things. So I have some ideas for in scope and out of scope. Um so I think these are some main points but if you see something that you would like to bring that we should add to this list uh of course feel free to to chime in. Um the first thing that I think we should discuss indeed is um what should be the functional boundaries with the other working groups. Um should we consider scheduling? Should we consider resource management as part of our analysis or should we focus only on the pulse level down sort of um part of the stack?

Michael Ferguson: I don't see why you would need to um um focus on scheduling. Is there a reason you you would argue for it?

Andrea Delgado: Yeah. Um so these questions my my my way of thinking was if we identified a set of use cases. um we run into like a phase space where we have like timing regimes in one of the axis and then u requirements in like um bandwidth and resource allocation.

00:21:13

Andrea Delgado: So the other question that sort of um sparked um whether we want to um include scheduling and and runtime was should we consider the use case of um cloud access for example? Is this something that we still want to consider or are we considering only on prim? Because if it's um a cloud access or perhaps a hybrid model where you want to run a CPU and a a QPU at the same time, but you want to time them where there's like a a very tight feedback between those two, you will also need to consider um resource management and scheduling. But perhaps that's too broad and we want to focus only on like the actual execution from the pulse level down.

Michael Ferguson: Yeah, I mean it sounds to me like that's something we we have for context and it's it's it's uh sort of broadly relevant for the description of the use cases, but I don't think that it's the domain of this working group. So probably shouldn't get much detail as far as far as cloud uh cloud systems. I think for the purposes of this working group may maybe we would assume that if you are doing any kind of um co-scheduled uh thing or you know real-time decoding or something like that that the the resource you are co-scheduling is nearby to the QPU maybe you're accessing them both through the cloud.

00:22:56

Michael Ferguson: Um that's what I I would advocate for here. Um,

Andrea Delgado: That's

Michael Ferguson: I think that the we don't really care that much about those like if it's uh uh if it's not if there's like a part of a workflow running you know nearby that or far away it doesn't matter that much for workflow stuff but for the any thing like real-time decoding or the fast feedback cases then I think we should assume that it's somehow next next to Go ahead.

Andrea Delgado: I completely agree. Um, and for the record, since we're taking notes, um, for the minutes, um, Matias also had like, um, a thumbs up. That's my personal

point of view as well that we should focus on on prem and and move away from from cloud. Uh Daniel

Daniel Weigand: Uh yeah, I had a slightly different point in in interface that's missing um namely I think an interface to the compiler group. So if I understood correctly, compiler still means gate level. um there will be a lot more compilation happening from physical gate level until you actually execute.

00:24:15

Daniel Weigand: Um and there you need to have good interfaces also for stuff like caching, partial compilation, these types of things.

Andrea Delgado: I will let Michael comment on that since he's the compiler group

Michael Ferguson: Yeah, I I mean one of um all of those things are are relevant and important

Andrea Delgado: expert.

Michael Ferguson: and one of the um aspects to consider is if there is compilation from gate level to pulse level can we do that um you know in uh what whatever environment is running the quantum SDK, does it have to happen in a service somewhere? Would it be possible to do it in a just in time way or does it need to be done ahead of time? Those are all questions that I think this working group could help answer. um where I I mean I have opinions about those things but um I think that it's kind of more about the um the control electronics uh picture what what is the result of compiling to pulse level is kind of u what what we would need to be defining in order to say anything about that um and we might say well we don't want to define that because that that's not possible to do in a community way.

00:25:41

Michael Ferguson: I don't know if we want to take an opinion on that um that but but uh from a sort of standardization point of view ideally we would have a community standard for what is a program running on the control electronics like in terms of you know what it what are the instructions and such. We just might not be at a place where it's time to make such a thing yet.

Andrea Delgado: I green that All right.

Santiago Nunez-Corrales: important. I want to go back to the point about scheduling. I agree with what has been said though it needs to be clarified. The clarification goes along the line of HPC that needs scheduling in the sense that we understand scheduling in HPC like slurm and similar versus HPC that needs to have immediate regular fast access for the purposes of supporting the micro architecture that the runtime is going to specify. I am not sure that distinction is clear and we would make a substantial service to the community by making that line very clearly drawn.

Andrea Delgado: Yeah. The way I was thinking about it, it was more of um what parts of or what um features of the control stack do we need to expose to make sure that the scheduling and whatever comes above pulse level is able to do like its job efficiently.

00:27:22

Santiago Nunez-Corrales: Correct. Uh I agree with that view. What I'm trying to say is that the at the end of the day events happening at the control level that require HPC will have different scheduling needs and different scheduling intensity. But those have to be encapsulated somehow as part of the micro architecture supporting the QPU operation rather than from the application point of view or program point of view. Something that has to be scheduled in the sense of slurm. There is a substantial distinction there. Again, we we seem to know that it's not explicit. So making that explicit may be worth thinking about. Sure.

Andrea Delgado: Okay. So then can I task you to write something about that?

Santiago Nunez-Corrales: I've been dying to write something like that for a while. So, yes.

Andrea Delgado: Done. Okay, that's great. Um, okay. So I think from the discussion that just happened, we're all in agreement of like batch workflow, co-ch schedule, um resource and capability discovery, uh control and data plane interaction, um including measurement streams, parameter updates, telemetry and error reporting.

00:28:36

Andrea Delgado: Is there anything else that you would like to see here in terms of information flow like in the bigger scope? Because again, we're going to bring back that data flow diagram that we started from and the idea is to convert that into um this set of interactions and data locations flows.

Santiago Nunez-Corrales: If I

Daniel Weigand: Um, I think one thing that came up that's not new in HPC but new to the quantum community is the the system admin stuff like do I have any hardware failures? Um, like the the monitoring of of the actual hardware

Andrea Delgado: Yeah, I think broadly that falls on telemetry as well.

Daniel Weigand: Yeah, I just want to make the distinction.

Andrea Delgado: the distinction.

Daniel Weigand: This is not about QPU but electronics.

Andrea Delgado: Yeah,

Daniel Weigand: So I don't know fans.

Andrea Delgado: something.

Santiago Nunez-Corrales: It'd be great to include data volume, program sizes, communication intensity because that defines bottlenecks.

Andrea Delgado: Okay. Yeah. Uh so I think that was the the next point but I agree. Um okay classical accelerators interfaces between uh the control box FPA GPUs CPUs um existing standards and protocols open source systems vendor approaches as many as we can and then sorry someone was going to say

00:30:28

Daniel Weigand: No, I just Sorry.

Andrea Delgado: And also again since Amir is here we're also thinking how to map this to the SC26 architecture but we're not going to use that as a reference. Um okay out of scope for milestone one of course endorsing a fair vendor software stack productuler um standardizing at this point again um the discussion right now is we're gathering information we're understanding bottlenecks where um things are right now but we're not doing any standardization as of now um defining device specific um interfaces and we're also going to try to be as cubit agnostic as possible. Um the idea again is not to publish um a comparative vendor ranking. This is just um um a discussion of what we're seeing. We're not trying to make an opinion. Um we're not thinking about bottlenecks suggestions as of now and we're not going to do uh precise numerical values. This is in scope for milestone number two. Uh some remaining questions if we agree on the scope. I think first one we have already discussed uh should we own the full path from resource management to QBU?

00:32:02

Andrea Delgado: I think we agree that that's out of scope and we should focus on on the runtime and control feedback boundaries. I see everyone saying yes by shaking heads. Um also um okay an important one should we consider the case for multiqpu and distributed quantum systems or not at this point or

Daniel Weigand: is it a case with entanglement or uh so because my my answer would be if there's entanglement yes but if no then it might might as well go by the the standard skinare at the

Andrea Delgado: I mean, we could either take both cases or or I think none. But do we have any insights in the group? Um perhaps air since you're already here are the other working groups considering uh distributed quantum computing either for like scheduling or like the runtime group. No answer. So I guess we go with Tara.

Santiago Nunez-Corrales: I'd say it should be included for distributed quantum systems. They are all what we have is relatively speculative. I agree with Daniel in the sense that if there's entanglement, there will be more complexity for ADHP system.

00:33:58

Santiago Nunez-Corrales: I do agree that we should do multiQPU even if it's if separate that should not be part of the scheduler in my experience that is that belongs to the micro architecture definitions.

Andrea Delgado: Yeah. Um, Michael

Michael Ferguson: I was just going to ask uh if we can predict whether or not it will impact any decisions we make. Um I'm inclined to think that it would in that um a multiqpu system is going to have more needs for things like the um fast feedback with some kind of remote. I mean even if we're just doing some uh like if we have quantum networking we have quantum networking with uh bell pairs then to do teleportation we have to actually move classical bits around too and then there's need for um low latency communication paths in and out of the control electronics. So I think it's worth I basically agreeing with Santiago that it is worth describing as a speculative use case. We explicitly say this is not a very mature area. Um but in broad strokes we that it will need the AB and C kind of functionality.

00:35:24

Amir: Can you guys hear me now?

Andrea Delgado: Yes,

Michael Ferguson: Yes.

Andrea Delgado: we can hear you now.

Amir: Sorry,

Andrea Delgado: But now you're after Matias because Matias is next.

Amir: I was first.

Matthias: Yeah. Uh my my connection is maybe a bit unstable, but uh originally I wanted to say that for me that feels like too speculative to to put it into into the white paper, but after Michael and Santiago's explanation, I think the way it was raised how to phrase it, it's it's perfectly fine and it's speculative and we see those requirements, but don't overengineer that section because it is speculative. So I'm fine. that I

Andrea Delgado: Come here.

Amir: Um so answering the first question we are looking at the embarrassingly embarrassingly parallel cases uh of multi- uh QPU the deployments in both the runtime and the resource management. So I guess I would go with the very first um I can't remember who said it but like the focus would be on entanglement rather than the the multiqP deployments on as long as there's no uh sort of tight communication loop between both of them.

00:36:35

Amir: I don't see why you would worry about it at the control level unless the control level is controlling both QPUs at the same time. Is that the what you guys are thinking about?

Andrea Delgado: Yeah. So what I gather from the discussion right now is we should make the distinction of we are considering distributed quantum computing in the case that it involves entanglement precisely because if it involves um entanglement that means that the control stack is actually responsible for managing this entanglement because you will have like the different control boxes like either entangling or sending information between the different cubits. But if this is um distributed in the sense that you're just sending the same job to different QPUs and you don't care if there's entanglement between them, then it's not in scope for our working group because it's more of like scheduling or like um higher levels of the stack.

Amir: Yeah, I agree. Uh the other thing I would add though is even with entanglements you will need interaction with the higher level resource management because something will have to uh configure the this disentanglement somehow.

00:37:50

Amir: So there'll have to be some sort of interface with the higher level of the software so that that the entanglement can be configured in order for the controller drone extend to do its job.

Andrea Delgado: So that's why um in the next topic is the use cases. Um I think another not to add another degree of freedom but or complexity entropy to this discussion but I think we should also um catalog the different use case in terms of relevance because there are some problems that are native to the control stack and then there are problems like scheduling for distributed that are adjacent to the control stack.

Amir: Yep.

Andrea Delgado: So we should also I guess mention that in in our discussion. Um Santiago

Santiago Nunez-Corrales: There is one possible use case in which even if we have distributed if we have parallelly embarrassingly parallel task across QPUs there may be needs to synchronize. So we should not discard the possibility that that synchronization will look as a precursor to having more powerful QPUs that have the possibility of being partitioned. That has been discussed in literature and makes substantial sense.

00:39:15

Amir: Do you mind uh mentioning a little bit more details on that?

Santiago Nunez-Corrales: Sure. So think of it, let's say you have multiple QPUs that can be made to act as a single one via circuit cutting and knitting as a precursor of having a larger QPU that you can put in place and remove that. But that is going to be very likely a pattern of scaling up and testing systems or testing applications that

require many cubit counts. But then you have to simulate that you have a single large QPU and then you need synchronization at a deeper level even if it is a soft realtime synchronization across the multiple QPUs.

Amir: All right, makes sense. Thank you.

Andrea Delgado: And also for some of these cases um you're able to either partition or make your circuit fit into a smaller QPU because you're able to do things like branching or like midcircuit measurement. So this falls right into the control stack and not like software.

Santiago Nunez-Corrales: It it's a way to simulate having a larger QPU by having many smaller ones that you can assume that they have similar synchronization properties and then they you reduce what's called heteroetasticity in the measurements that you produce or the errors behind the measurements.

00:40:56

Andrea Delgado: Okay. So, I'm glad to hear we're all on the same page and reaching consensus. So, moving on to another question. Should we in this version in this iteration of um milestone one should we consider a security tendency and all of these different requirements of an HPC facility scale accelerator supercomput or is it too early to have this discussion? Michael,

Michael Ferguson: These details seem uh sort of more fine grained than the level that we are talking about working um in in milestone one. So I think probably leave them out.

Andrea Delgado: that's

Matthias: The only security feature which I would mention and and which should be there is that you make sure that the user cannot kill the QP with the instruction they sent to it. Right? that you cannot just fry the the device with with the pulses you you sent there. But besides that, I I agree with Michael that that is too early to any anything else.

Daniel Weigand: I think even that might be hard to do depending on cubid architecture. Some of them just like to blow up.

00:42:38

Andrea Delgado: Okay. And the last question, what terminology needs alignment with other open QSC groups? Um, and this is among the likes of a discussion that we were having last week at Quantum Week. Um, there are things like compilation that mean different things for different working groups. Um I think we should also agree on like timing domains. What does it actually means like real time near time? Um these are the ones that I can think of but is there any of these that we should think about. Um, starting with Matias

Matthias: Oh, no. This was this was the earlier hand race. I just put it down. Sorry.

Andrea Delgado: here.

Amir: uh just to put a plug in there. So the system work working group uh the system architecture working group is working on a white paper to basically define this. So it might be good to kind of join forces on this one. Uh um instead of having like two separate things trying to to define terminology, let's just have one output that defines terminology for the entire open QC initiative.

00:43:53

Andrea Delgado: Yeah, that's why I wrote it as alignment and not defining because I know that we are um later in the game than other working groups. But who should be the person to contact?

Amir: Uh so the two people that are running this is Johannes Belchuk and um uh Sebastian Stern. I'll anyway I'll I'll send out the contact info and maybe we can get one of the representative from here to join that working group and somebody from that working group to join this one so that there's like whenever this point is discussed this way there's a bit more cohesion

Andrea Delgado: Okay, done.

Daniel Weigand: Um just to add a few more things to the list. Uh I also noticed in this document and and it was a hardware vendor uh QPU. Uh sometimes QPU means the entire control stack including the uh electronics. Um

Michael Ferguson: Yeah, that's one of those things it'd be great to just have what what are we in and open QSC going to do? Just pick one so we can stop confusing each other. Hey.

00:45:12

Andrea Delgado: Yeah, you guys were not there, but I think this was one of the first points that Santiago brought up. Are we saying the QPU is the the cubits and in control stack or is it like the whole thing?

Santiago Nunez-Corrales: small suggestion maybe food for thought not for any decision maybe the QPU is the layer that the compiler interfaces with. Now that unravels a whole rabbit hole, right? But we need to think more deeply about this.

Andrea Delgado: I'm sorry.

Jack Lange: Yeah. Well, so I have a related question which is like how do we classify the control stack, right? So the TPU in the control system and then you have all the QC decoders on the back end. I'm wondering how far the control system reaches into the QC decoding infrastructure. Um so I can imagine like with some HPC resources, you know, it's much looser um loosely integrated uh than we would think standard control systems. Um I guess the one proposal I would have I think you mentioned the scheduling regime. um seems like there might be a boundary there where we could classify the control system as simply as something that HPC res or infrastructure that's scheduled with the QPU for a given job.

00:46:50

Jack Lange: Um and things that aren't scheduled independently might be a decent demarcation for thinking about that.

Michael Ferguson: Yeah, along those lines,

Andrea Delgado: Yeah.

Michael Ferguson: like at HP, we've been thinking a fair amount about um sort of midcircuit classical and the spectrum of where you might want to run that. And I think there are use cases for doing that in an HPC system, but it would only make sense if you co-scheduled, right? That kind of thing. You it is nonsense if you didn't have both resources ready to go. Um I think most of the most of that pattern in practice is probably going to happen with uh with some dedicated hardware that just always gets allocated with the QPU. you might not think of it as an HPC node. You might think of it as you know the QPU accelerator or something. But um I think there's there's need for both in in terms of capability. Um but the the difference is it it depending on how you build it might not be that diff that much right because um if you have the ability to go out of the control electronics over a network then you could go to an HPC node or something sitting you know in the same same rack as the control electronics either way.

00:48:18

Michael Ferguson: Um, but some of the systems maybe they don't go over the network. Maybe it's like more of a uh, you know, an ARM CPU that's sitting next to the FPGA and there's some faster path that I wouldn't be able to summarize, but I I bet our QLocks friends have a better idea what that kind of thing looks like. Um, so those those are very different in terms of what kinds of uh things we would specify. Um but from my point of view I think there's a need to do both of those things in terms of describe a path over the network for uh midcircuit operation and also a path for sort of more uh tightly integrated hardware.

Jack Lange: And could we define the control system as the nonquantum infrastructure that needs to be co-scheduled with the quantum infrastructure.

Andrea Delgado: I think that's a great way to

Michael Ferguson: I I'd probably I'd probably add to that that it's it's dedicated reserved sort of system level hardware that we consider associated with the the QPU because like I was saying there are cases where you might want to have a co-scheduled HPC node doing something Maybe you're doing a QEC experiment and you need different kind of hardware or something like that.

00:49:38

Michael Ferguson: Um I think those are cases but we need necessarily to go over the network and we wouldn't consider those HPC nodes part of the QPU. Um if we use that term

Andrea Delgado: What's that one?

Santiago Nunez-Corrales: Maybe we need some something a bit more abstract that that seems to be too much in the wits of how we do things now or how we can imagine them. We may be mistaken an failure of imagination of an insight into necessity. A control system is probably defi, we could define it as anything any piece of machinery that needs to be connected to satisfy what the runtime should do. the runtime meaning that we are able to take instructions that are symbolic and render them as something that goes into hardware that is into the specific actions that render what the instructions specify. For me the boundary is between symbolic and something that happens that is mechanical that that is robotics. how we do it whether it is HPC it is not HPC FPGA that is a matter of circumstantial convenience but I think that if we depart from the sequence of events I remember Andrea's diagram there's a point in which we get symbolic information and then that is that has to be executed as um the deeper layers of of hardware synthesis I'm not going to call it compilation to not to confuse it down to pulses down to the whole dispatching of events.

00:51:23

Santiago Nunez-Corrales: I'm not going to use orchestration again to to avoid that. That's why probably we need to craft this definition and and hammer it out between each other. But the symbolic layer, we have a program, that program goes into the quantum uh resource management interface. There's a request for resources. Those are signals. Those are symbolic. But the moment that a program reaches the level of the runtime, this is when it meets the metal, it meets the cubits. How that happens is the responsibility of the quantum control system. It's like a control unit in a microprocessor. At least that's how we've been thinking about it.

Michael Ferguson: So Santiago in that case the kind of scenario I was describing where if we have an HPC node assisting with some midcircuit thing you would just say well in that specific job we've sort of created a virtualized part of a a control system that that is that HPC node. So you would say it is part of the control system. It's just it's not uh fixed into the hardware that it's part of the control system.

00:52:30

Santiago Nunez-Corrales: Exactly the the conse the byproduct that it forces a different sort of access pattern for the HPC is purely circumstantial. It will eat up eat away a bit of the HPC system for the fact that we need quantum control which is separate from executing applications. That's why that I think that needs to be said but the specific instancing is by virtue of of of the needs that we have right now. Maybe somebody comes up with a better let's say cryogenic system that is embedded directly into the QPU and and excuse me for not saying what a QPU is right now. But let's say that's this box that is semi that has autonomy of control that we send signals to and symbolic signals or symbolic programs to. But yes, I agree with with how you put it. virtualized micro architecture of sorts.

Andrea Delgado: My problem with treating it as um just a thing where we said like pulses it is not um quite the case. So um I brought up like the diagram that that we started the discussion at the inerson meeting.

00:53:40

Andrea Delgado: So everything here on the gray box is basically the control box like everything outside is like compilation and all of that but inside the control box you also have um computing power because at the end what comes out of the box is not like the actual polls that you get from readout but is it is actually a process set of data. So I I wouldn't treat it as just something that you send pulses and and get pulses out. Like there is some computation, some information processing that happens inside the box. Um Michael

Michael Ferguson: Yeah. I mean, I think what Santiago was saying is that we give some kind of symbolic description of what we want to do to the cubits. We hand that to the the the thing that we call the QPU that or whatever name we come up with that includes the control electronics and any helping CPUs or GPUs or whatever and then that that is the responsibility of the CPU to do in as efficient or whatever we call it that box to do including the control electronics to do in as efficient a manner as possible which there are many ways to handle.

00:55:02

Michael Ferguson: It might involve ahead of time converting the program into a pulse level program or involve um doing you know FTQC primitives uh one at a time um by sort of playing a a program for each one. Um there there's a lot of a lot of possibilities there. But I like Santiago's framing of well, we've we've got a an idea of what a circuit level program is. We hand that the the QPQ is the thing that that runs this what we call a circuit level program, right?

Andrea Delgado: But then where would where would the decoder fall?

Michael Ferguson: Um and then go ahead.

Andrea Delgado: Would they be part of like the computer?

Michael Ferguson: It's part of the in in that framing. It's part of the Q. Well, I don't know if we call it the QPU, but we're going to describe this box. We'll give some name

to it and it would include the decoder. We might not call it QPU. We might call it something else.

Santiago Nunez-Corrales: So what I'm trying to get at Yeah,

00:56:03

Andrea Delgado: Hello city.

Santiago Nunez-Corrales: it's exactly what what what we are trying to say is that the word control needs to grow up as we grow up these architectures. What we mean by quantum control today is really pulse level control but that is probably at a lower level. there there's a higher level that is event control of the system and probably this also mandates that there will be something that is a QEC box I don't know even how to call it I'm going to use Michael's terminology box period we have a QC box that has signals has data and can be controlled by our larger control unit and then we start to see the first time and I'm excited this is why this is so exciting for me we are starting to see the birth of quantum computer organization properly. We don't have that yet. And this is exciting.

Andrea Delgado: having dinner.

Daniel Weigand: Yeah, that that was actually briefly what I thought we basically just describing a quantum computer as a just whole system and then you have basically everything else that just submits a job and then the The quantum computer whatever we actually pack in there just executes the thing.

00:57:25

Andrea Delgado: Oh, I'm done.

Jack Lange: Um yeah I guess so the one issue I see with that formalization of looking as you know it's essentially the the interface is the instruction or circuit level representation. Um, I think that broadens out certainly the interfaces we're talking

about much deeper into what that representation is because you're going to have to capture things like QEC um, approaches. Um, there's a lot of additional information I feel like that's going to have to come down with that beyond just the circuit representation. So suddenly we're now having to interface I guess with the compiler teams and all of that as opposed to figure out how do we just connect you know these boxes together um inside our kind of context. So I guess that gives me a little bit of concern of how the scope creep of sort of you know is that really what we want to be doing is defining like what the that interface is as far as you know compiler is going to give us everything we need and what is that going to be. Um or do we want to focus more on just how do we integrate things is the question

00:58:34

Daniel Weigand: Um maybe a question from my side is uh if I if I think about very

Andrea Delgado: Yes.

Daniel Weigand: big cubes like thousands of cubits then that information has to be passed along to the electronics anyway because we wouldn't be able to store that program. So, so the the whole control flow,

Andrea Delgado: Thanks.

Daniel Weigand: the the QC information, all of that would need to go as basically one package. Um, so in that sense, it's a bit of a we don't really have a choice or there can be another work group that does it,

Jack Lange: No,

Daniel Weigand: right? But it needs to happen at some point.

Jack Lange: I guess is the question. Do we want to define that or is it more are we trying to get a handle on the requirements for it? Um

Andrea Delgado: so this uh stage we don't want to define anything um since it is the top of the hour and I want to respect um everyone's time. So the next part of the agenda was to discuss use cases.

00:59:49

Andrea Delgado: So I will task you with a bit of a homework and I'll share with you like the table for the use cases and if you don't mind spending some time over the next couple of days just like voting for use cases arguing for whether or not we should include it so that um by next Wednesday I can probably compile that information and then take volunteers if you want to dive deeper into those use cases and and requirements. I'll share a template so that we're all working under like the same base of questions and and it's easier also to compile that information. So in the meeting in two weeks we can move to um a different topic and agree on these use cases. Um I echo uh Santiago's um message about saying that this is very exciting. Um, I'm happy to see that we're all like engaging into the conversation and and I'm happy to see where this goes. Michael

Michael Ferguson: sort of a triviality, but the uh notes you've been sharing um are not in the um the main uh um fork of working group controls. They're in your yours in a particular branch. Are you planning to push them to the main one so it'll be easier for us to see them?

Andrea Delgado: as soon as the boss approves them. I already did like the pull request.

Amir: I just gave you I I just gave you admin access,

Andrea Delgado: So we'll do

Amir: so you should be able to push it.

Michael Ferguson: Okay, thanks.

Andrea Delgado: and I will share the GitHub and if you're not um able to access let well let me or Amir know because he's the boss and we can give you access. Um and other than that I think Slack is the best way to also discuss asynchronously and I guess um I'll also post the notes of like today's discussion like it's being transcribed and summarized and everything. So I'll post and I guess I'll see you in two weeks.

Michael Ferguson: Thanks.

Amir: All right.

Daniel Haag: Thank you.

Andrea Delgado: Bye.

Amir: Thank you.

Santiago Nunez-Corrales: All right.

Daniel Weigand: All right.

Matthias: So, uh,

Transcription ended after 01:02:15

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